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Course: CSA0421 - Operating System | **GitHub Repo:** krishshiva721/CSA04--Assignment1

TASK 4: OPERATING SYSTEM SERVICES

Service Allocation, Minimalist Stripping, and Security Isolation

1. SERVICE ALLOCATION & STRIPPING MATRIX

OS Service Category	Bedside Monitor Status	Database Server Status	Technical Justification
User Interface (UI / CLI)	Stripped / Stripped	Minimal CLI / GUI Disabled	Monitors present raw framebuffer signals; stripping shell/window managers frees memory and eliminates security vectors.
Program Execution	Static Single Tasking	Dynamic Multi-threading	Monitors run pre-compiled fixed tasks. Servers require dynamic process spawns for connection pools.
File System Manipulation	Stripped (Raw RAM Log)	Essential (Journaled POSIX)	Monitors avoid disk writes to eliminate storage IO latency. Database requires transactional ACID filesystem logs.
Protection & Security	Hard Memory Protection	RBAC & ACL Enforced	Database OS isolates query sessions by user IDs and virtual address spaces to prevent cross-patient record leaks.
Accounting & Auditing	Minimal Local Counters	Essential System Audit Log	HIPAA regulations require mandatory compliance logging of every database record modification.

2. PATIENT RECORD ISOLATION ARCHITECTURE

The database OS enforces isolation using **Discretionary Access Control (DAC)**, **Mandatory Access Control (MAC/SELinux)**, and **Virtual Memory Address Space Separation**. Each client request handler executes under a distinct UID/GID context. Page tables isolate physical memory frame allocations, ensuring Process A cannot inspect memory pages of Process B, preventing unauthorized memory probing.

3. ALTERNATIVE CONCEPT & COMPLEXITY ANALYSIS

Alternative Scenario: Handheld Tablets Running Full Desktop OS vs. Optimized Mobile OS

If clinical tablets were configured with a full desktop OS (enabling background indexers, legacy daemons, print spoolers, and unthrottled desktop environments):

- **Space Complexity (RAM & Storage):** Baseline OS RAM usage grows from 1.2 GB (Mobile Android/iOS base) to 4.5 GB+ (Full Desktop OS). Storage overhead increases by 35 GB for unused drivers and binaries.
- **Time Complexity & Boot Overhead:** Cold boot time increases from $T_{boot} \approx 12 \text{ s}$ to $T_{boot} \approx 45 \text{ s}$. Background daemon scheduling increases idle CPU wakeups, reducing battery shift duration from 14 hours down to 4.2 hours (a 70% decrease).